

Towards AI That Understands the Human World

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Abstract

AI has reached a turning point. Systems can now perceive, generate, and act in language and image across digital platforms at unprecedented scale. Yet as AI moves from tools to collaborators—embedded in decision-making, institutions, and everyday life—a new requirement becomes unavoidable: AI must understand the world the way humans inhabit it.

This talk introduces Cognitive World Modeling as the next phase of AI development. It unifies physical world modeling—time, space, causality, action—with mental world modeling—goals, beliefs, intentions, emotions, and social norms—into a single, persistent representation of reality as experienced by humans. Together, these models allow AI systems not only to predict outcomes, but to reason about meaning, context, and consequence. Cognitive World Modeling moves AI beyond reactive toward systems that can plan, explain, adapt, and collaborate over time. Alignment and trust emerge not as post hoc constraints, but as properties of systems that maintain accurate, evolving models of both the external world and the humans within it.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

Cognitive world modeling, human-centered AI, theory of mind, multimodal reasoning, embodied agents, AI alignment, AI ethics

BIOGRAPHY

Pascale Fung is a Fellow of the ACL, AAAI, IEEE, and ISCA for her significant contribution to human-machine interactions and AI ethics. She is an expert on the Global Future Council of the World Economic Forum since 2016. She is on the Expert Network of the UN Advisory Body on AI. She is a Chair Professor at the Department of Electronic & Computer Engineering at The Hong Kong University of Science & Technology (HKUST), and a visiting professor at the Central Academy of Fine Arts in Beijing. She previously led the Embodied AI Agents research effort at Meta-FAIR.



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WWW '26, April 13-17, 2026, Dubai, United Arab Emirates.

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ACM ISBN 979-8-4007-2307-0/2026/04.

<https://doi.org/10.1145/3774904.3795890>